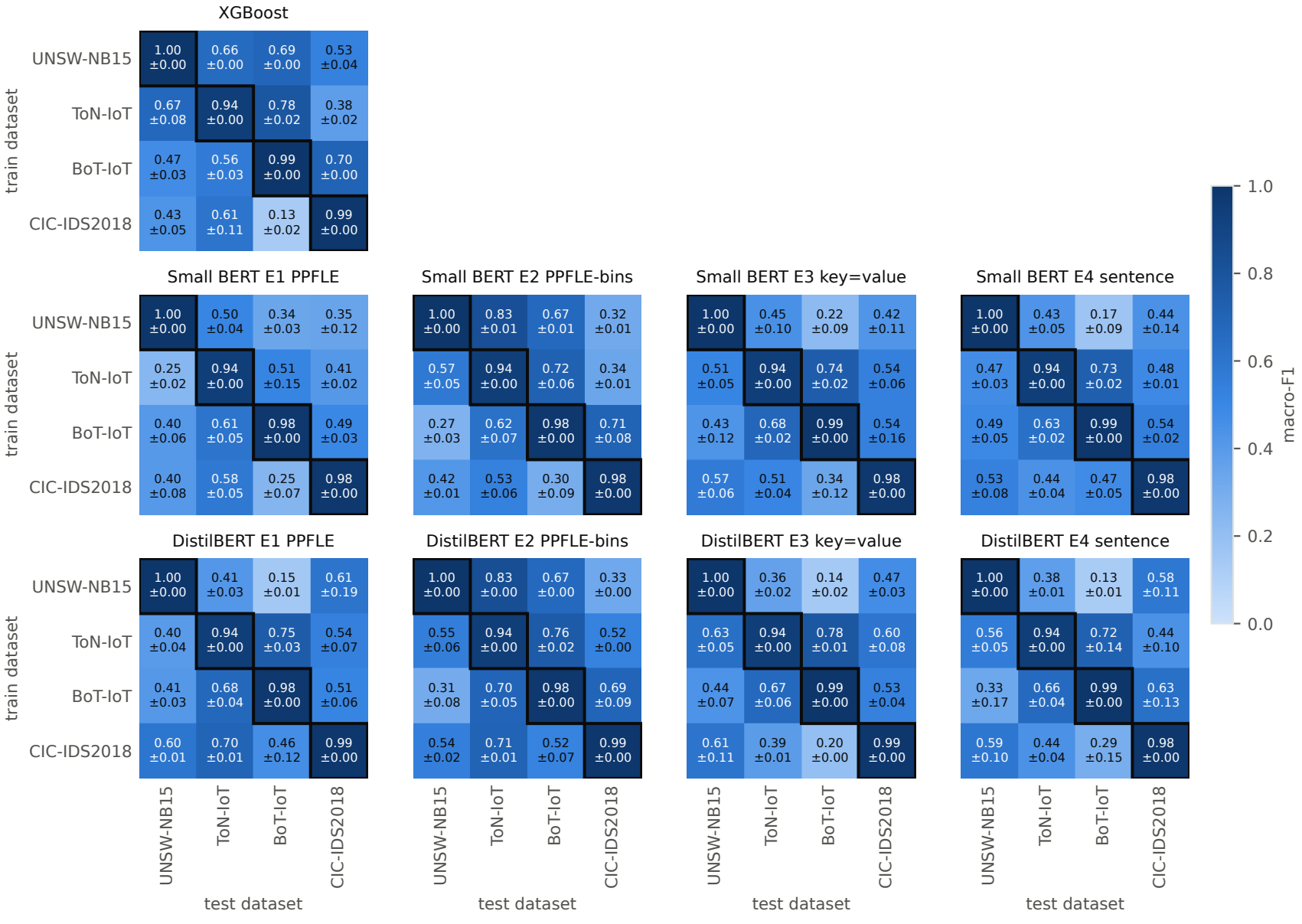
XGBoost
Small BERT E1 PPFLE
Small BERT E2 PPFLE-bins
Small BERT E3 key=value
Small BERT E4 sentence
DistilBERT E1 PPFLE
DistilBERT E2 PPFLE-bins
DistilBERT E3 key=value
DistilBERT E4 sentence
train dataset
UNSW-NB15
ToN-IoT
BoT-IoT
CIC-IDS2018
test dataset
macro-F1
1.0
0.8
0.6
0.4
0.2
0.0
1.00 ±0.00 0.66 ±0.00 0.69 ±0.00 0.53 ±0.04
0.67 ±0.08 0.94 ±0.00 0.78 ±0.02 0.38 ±0.02
0.47 ±0.03 0.56 ±0.03 0.99 ±0.00 0.70 ±0.00
0.43 ±0.05 0.61 ±0.11 0.13 ±0.02 0.99 ±0.00
1.00 ±0.00 0.50 ±0.04 0.34 ±0.03 0.35 ±0.12
0.25 ±0.02 0.94 ±0.00 0.51 ±0.15 0.41 ±0.02
0.40 ±0.06 0.61 ±0.05 0.98 ±0.00 0.49 ±0.03
0.40 ±0.08 0.58 ±0.05 0.25 ±0.07 0.98 ±0.00
1.00 ±0.00 0.83 ±0.01 0.67 ±0.01 0.32 ±0.01
0.57 ±0.05 0.94 ±0.00 0.72 ±0.06 0.34 ±0.01
0.27 ±0.03 0.62 ±0.07 0.98 ±0.00 0.71 ±0.08
0.42 ±0.01 0.53 ±0.06 0.30 ±0.09 0.98 ±0.00
1.00 ±0.00 0.45 ±0.10 0.22 ±0.09 0.42 ±0.11
0.51 ±0.05 0.94 ±0.00 0.74 ±0.02 0.54 ±0.06
0.43 ±0.12 0.68 ±0.02 0.99 ±0.00 0.54 ±0.16
0.57 ±0.06 0.51 ±0.04 0.34 ±0.12 0.98 ±0.00
1.00 ±0.00 0.43 ±0.05 0.17 ±0.09 0.44 ±0.14
0.47 ±0.03 0.94 ±0.00 0.73 ±0.02 0.48 ±0.01
0.49 ±0.05 0.63 ±0.02 0.99 ±0.00 0.54 ±0.02
0.53 ±0.08 0.44 ±0.04 0.47 ±0.05 0.98 ±0.00
1.00 ±0.00 0.41 ±0.03 0.15 ±0.01 0.61 ±0.19
0.40 ±0.04 0.94 ±0.00 0.75 ±0.03 0.54 ±0.07
0.41 ±0.03 0.68 ±0.04 0.98 ±0.00 0.51 ±0.06
0.60 ±0.01 0.70 ±0.01 0.46 ±0.12 0.99 ±0.00
1.00 ±0.00 0.83 ±0.00 0.67 ±0.00 0.33 ±0.00
0.55 ±0.06 0.94 ±0.00 0.76 ±0.02 0.52 ±0.00
0.31 ±0.08 0.70 ±0.05 0.98 ±0.00 0.69 ±0.09
0.54 ±0.02 0.71 ±0.01 0.52 ±0.07 0.99 ±0.00
1.00 ±0.00 0.36 ±0.02 0.14 ±0.02 0.47 ±0.03
0.63 ±0.05 0.94 ±0.00 0.78 ±0.01 0.60 ±0.08
0.44 ±0.07 0.67 ±0.06 0.99 ±0.00 0.53 ±0.04
0.61 ±0.11 0.39 ±0.01 0.20 ±0.00 0.99 ±0.00
1.00 ±0.00 0.38 ±0.01 0.13 ±0.01 0.58 ±0.11
0.56 ±0.05 0.94 ±0.00 0.72 ±0.14 0.44 ±0.10
0.33 ±0.17 0.66 ±0.04 0.99 ±0.00 0.63 ±0.13
0.59 ±0.10 0.44 ±0.04 0.29 ±0.15 0.98 ±0.00